Physiological Constraints and Behavioral Adaptation: How high stakes professionals respond to fasting during Ramadan

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October 22, 2026

Abstract

Physiological constraints are often assumed to reduce worker productivity, but workers may instead adapt by changing effort across tasks or over time. We study this question in the Indian Premier League, a high-stakes professional cricket setting with ball-by-ball measures of output. We examine how Muslim batters respond to Ramadan fasting, a predictable physiological shock that potentially affects production. We compare Muslim and non-Muslim players using a difference-in-regression-discontinuities design that uses rule changes over the course of IPL games during and outside the Ramadan period. Muslim players during Ramadan smooth performance over match phases: early high-intensity production is muted, while later production is preserved or increases. Complementary difference-in-differences estimators suggest similar within player results and that teams respond modestly by substituting fasting players at higher rates. The results suggest that when production is divisible and workers can substitute effort across tasks, predictable physiological constraints can alter the composition of output more than its level.

1 Introduction

Physiological and health shocks are typically evaluated through their effects on average output. Such estimates reveal whether productivity falls, but not how workers adjust the timing or composition of effort (Smith, 1999). Becker (1977) treats effort as a scarce input that workers allocate across productive uses, while Holmstrom and Milgrom (1991) emphasize that effort devoted to one productive margin affects the amount available on others. Together, these frameworks imply that an increase in the marginal cost of exertion may induce workers to reallocate effort across tasks or over time, potentially preserving aggregate output even as patterns of production change. However, this adjustment margin is rarely observed because effort is difficult to measure directly and most production settings do not divide work into clearly separable phases.

We study this question in the Indian Premier League (IPL), a professional sports setting in which individual output is measured at high frequency and incentives are strong. The IPL is one of the most lucrative and popular sports leagues in the world with over 462 million TV viewers and estimated worth of $6.8 billion (Wells and Weinstock, 2020). This environment provides an opportunity to examine whether workers facing a predictable physiological constraint adjust how effort is allocated within production. In particular, we examine how fasting during the Muslim holy month of Ramadan affects in-game performance among Muslim players and whether they adapt their playing strategy to avoid productivity declines.

A distinctive feature of IPL Twenty20 cricket is that production is segmented into discrete phases with different returns to effort. During the opening six overs of play (the "powerplay") opposing-team fielding restrictions incentivize aggressive, high-exertion batting play. After the powerplay, returns to effort fall mechanically as fielding restrictions are relaxed, and scoring rates typically decline (Silva et al., 2015). This rule-based structure creates a setting in which to study time-varying effort allocation when workers are subject to a health shock.

We combine this powerplay cutoff with Ramadan fasting as a predictable physiological constraint. Our empirical design compares Muslim players during Ramadan to Muslim players outside Ramadan and to non-Muslim players in both periods. On average, batting productivity drops discontinuously as the innings transition from the powerplay to the middle overs. The exception is that Muslim players during Ramadan do not exhibit this drop. Instead, they smooth performance across match phases, reducing productivity in the power-

play overs while maintaining or increasing productivity in the later middle overs. Consistent with fasting being the main mechanism, effects are larger during the day vs at night when fasting is broken after sunset.

We use the onset of Ramadan and its end to estimate complementary effects via DiD estimators comparing Muslim and non-Muslim players. We find evidence supportive of this change in play. Declining runs per ball in the powerplay are offset by increasing runs per ball in the middle and death overs. Declines in production in the powerplay are offset by increases in the middle and death overs for Muslim players. Our headline DiD results suggests preservation of runs per ball over the course of games.

This preservation of within-game player productivity also results in muted team organizational responses. We find some evidence that Muslim players are substituted out of team line-ups during a Ramadan game with higher frequencies. We also find clinically modest evidence that Muslim players are moved up in batting order by about 0.3 places during Ramadan. Finally, we find statistically marginal evidence that when able, teams choose to bat second in afternoon games when fasting is ongoing. However, we find no evidence of adverse labour market consequences in terms of employment or compensation. We estimate a null effect on overall team runs and probability of team wins although these regressions are noisy and we cannot rule out moderate negative effects.

These findings speak to a broader literature that treats effort as a scarce input allocated across productive uses. Becker (1977) emphasizes that workers choose the intensity of effort as well as time supplied, while Holmstrom and Milgrom (1991) highlight that effort devoted to one productive margin can affect the amount available on others. In our setting, a predictable physiological constraint appears to alter this allocation: Muslim batters mute high-effort production during the powerplay while preserving lower-effort production later in the middle and death innings, leaving aggregate output largely unchanged. Because the return to aggressive play changes discretely at the powerplay boundary and output is measured ball by ball, T20 cricket makes this within-task reallocation of production unusually visible.

Our results also connect to a broader literature emphasizing that physiological capacity is an economically relevant and endogenously managed input into production. Prior work shows that workers adjust effort and time allocation in response to fatigue-related constraints, such as sleep deprivation, with productivity effects depending on behavioral responses (Bid-

dle and Hamermesh, 1990; Gibson and Shrader, 2018). Empirical evidence also demonstrates that labor supply is elastic along the effort margin even at very short horizons, with workers adjusting intensity in response to predictable conditions (Oettinger, 1999). Our closest analogue paper documents the effects of a health intervention on worker effort as measured by accelerometers. Akogun et al. (2021) show that a health intervention produced higher productivity workers with the mechanism being substitution between low to high effort tasks.

A large literature uses Ramadan fasting as a natural experiment to study the economic effects of religious observance, often focusing on average impacts on productivity, human capital, or labor market outcomes (Almond and Mazumder, 2011; Almond et al., 2015; Hornung et al., 2023; Bertoli and Grembi, 2024; Majid, 2015). In contrast to this work, we document how workers adapt behavior within production. Our results are most similar to Hu and Wang (2019) who show effects of Ramadan on cashier productivity in Indonesia using high frequency retail data.

Finally, while sports are frequently used as laboratories for labor market analysis (Kahn, 2000), existing studies of health shocks in professional sports largely emphasize performance losses following injuries or illness (Hayes et al., 2025; Chikish and Humphreys, 2025; Bullock et al., 2022; Fischer et al., 2022) with exceptions specific to Ramadan fasting (Abaïdia et al., 2020). Our findings highlight a complementary mechanism: when constraints are predictable and substitution is feasible, workers and organizations can reoptimize effort allocation to maintain productivity.

The remainder of the paper is organised as follows: Section 2 introduces a model that provides intuition on how a physiological shock such as fasting during Ramadan might affect production of runs in batters in IPL T20 cricket. Section 3 describes data, empirical methods and highlights key identification assumptions. Section 4 summarizes the main results. Section 5 concludes.

2 Model

We introduce a Grossman-style model of health capital and depreciation to explain how batters produce runs in IPL T20 cricket and how health shocks, like that of fasting during Ramadan, might affect production (Grossman, 1972). The goal of this model is to explain how our empirical results are consistent with mechanisms. Batters have a stock of non-

renewable health capital at the beginning of each game of H_0. Batters face bowlers who throw balls and so expend energy e_t in order to hit these balls and produce runs r_t. The goal of the batter is to maximize total expected runs. When facing ball t, the law of motion for a batter's health is

$$H_{t+1} = (1 - \delta)H_t - g(e_t). \tag{1}$$

There is passive depreciation in health that occurs as a result of continuing to bat, represented by δ, where $0 \leq \delta < 1$. The choice of the batter is what amount of energy e_t to expend for ball t. Effort depletes health capital according to $g(e_t)$, where

$$e_t \geq 0, \quad g'(e_t) > 0, \quad g''(e_t) > 0. \tag{2}$$

Thus, effort produces additional runs but with accelerating health cost. When the player exhausts their stock of health capital, they cannot produce further runs. The batter's production of runs is also a function of effort at ball t, where

$$r_t = A_t f(e_t), \tag{3}$$

and

$$f'(e_t) > 0, \quad f''(e_t) < 0. \tag{4}$$

Effort raises expected runs but with diminishing returns. Expected runs are also related to the batting environment, which is meant to capture the different phases of the game. These two environments are the powerplay, when there are restrictions on fielders playing in the outfield, and the middle overs, when fielders are able to play more freely in the outfield:

$$A_t = \begin{cases} A_{PP} & \text{if ball } t \text{ occurs during the powerplay,} \\ A_M & \text{if ball } t \text{ occurs during the middle overs.} \end{cases}$$

where

$$A_{PP} > A_M. \tag{5}$$

The batter's full dynamic problem is to choose an effort path,

$$\max_{\{e_t\}_{t=1}^T} \sum_{t=1}^{T} A_t f(e_t), \tag{6}$$

subject to

$$H_{t+1} = (1 - \delta)H_t - g(e_t), \tag{7}$$

with

$$H_t \geq 0, \quad e_t \geq 0. \tag{8}$$

To build intuition, first consider the benchmark case with no passive depreciation, $\delta = 0$. In that case, the dynamic constraint collapses to

$$\sum_{t=1}^{T} g(e_t) \leq H_0. \tag{9}$$

The Lagrangian for this benchmark case is then

$$L = \sum_{t=1}^{T} A_t f(e_t) + \lambda \left(H_0 - \sum_{t=1}^{T} g(e_t) \right). \tag{10}$$

The first order condition for each ball is then

$$A_t f'(e_t) = \lambda g'(e_t). \tag{11}$$

We assume the health-capital constraint binds, ie $\lambda > 0$; a batter for whom it is slack simply sets effort at the phase-specific optimum and exhibits no reallocation. The left-hand side of this condition is the marginal benefit of effort on ball t. The right-hand side is the marginal health-capital cost of effort. Since $A_{PP} > A_M$, the marginal benefit of effort is higher in the powerplay than in the middle overs. Because $f''(e_t) < 0$, the batter can allocate more effort to powerplay balls before the marginal benefit of effort falls to the common shadow cost of health capital. Thus, in the absence of health shocks, the model predicts $e_{PP} > e_M$: batters should optimally spend more health capital in the powerplay, when the return to aggressive effort is higher.

2.1 Health shocks

We posit three avenues where a health shock like fasting during Ramadan can influence how players can produce runs. The first avenue is through its direct effect on the initial stock of health capital, where

$$H_0^R < H_0. \tag{12}$$

A reduction in the initial health stock tightens the health-capital constraint. In the benchmark model, this raises the shadow value of health capital:

$$\lambda_R > \lambda. \tag{13}$$

The first-order condition becomes

$$A_t f'(e_t^R) = \lambda_R g'(e_t^R). \tag{14}$$

Since the marginal cost of effort is higher in all phases, optimal effort falls in both the powerplay and the middle overs. Because expected runs are increasing in effort, this implies

$$r_t^R = A_t f(e_t^R) < A_t f(e_t) = r_t. \tag{15}$$

A health shock that directly affects the capital stock should therefore cause a generalized decline in runs across all phases, while retaining the same incentive to expend more energy and produce more runs during the powerplay relative to the middle overs.

The second avenue where a health shock can influence batter production of runs is through an increase in the potential depreciation of health capital. This is equivalent to

$$\delta_R > \delta. \tag{16}$$

In the law of motion,

$$H_{t+1} = (1 - \delta)H_t - g(e_t), \tag{17}$$

a higher value of δ means that health capital erodes more rapidly over the innings, even holding effort fixed. This avenue predicts a time gradient in performance. Early in the innings, before much health capital has depreciated, production may be relatively preserved:

$$r_{PP}^R \approx r_{PP}. \tag{18}$$

However, as the innings progresses, faster depreciation should reduce available health capital and lower later production:

$$r_M^R < r_M. \tag{19}$$

This avenue predicts relatively normal early production followed by worse production later in the innings.

Finally, the health shock can raise the health cost of effort $g(e)$. The base FOC implies that this should result in effort falling. However, the scale of the fall depends on whether this increase in health cost affects high effort, low effort, or both. If Ramadan only raises the cost of high-intensity effort, then:

$$g'_R(e) \approx g'(e) \text{ for low } e, \tag{20}$$

and

$$g'_R(e) > g'(e) \text{ for high } e. \tag{21}$$

The first-order condition becomes

$$A_t f'(e_t^R) = \lambda g'_R(e_t^R). \tag{22}$$

Because middle-over batting typically requires lower effort, the Ramadan cost shock may have little effect in the middle overs. This in turn implies that

$$e_M^R \approx e_M \tag{23}$$

and

$$r_M^R \approx r_M. \tag{24}$$

However, powerplay batting normally requires higher effort. At these higher effort levels, $g'_R(e) > g'(e)$, so optimal powerplay effort falls:

$$e_{PP}^R < e_{PP}, \tag{25}$$

which implies

$$r_{PP}^R < r_{PP}. \tag{26}$$

During Ramadan, if the effective marginal cost of effort is high enough, this could potentially reduce run production in the powerplay to middle-over levels. This final channel is distinct from both a lower initial health stock and faster depreciation: it predicts relatively normal middle-over production but muted responsiveness to the high-return, high-effort opportunity created by the powerplay. Our results are mostly consistent with this third avenue where a health shock differentially affects worker effort.

3 Methods

3.1 Indian Premier League Cricket

Indian Premier League (IPL) Twenty20 (T20) matches consists of two innings of up to 20 overs with six ball deliveries per over with teams alternating between batting and fielding. During play, two batters face eleven fielders from the opposing team. Batters score runs by hitting the ball with a bat and running between the wickets or by hitting "boundaries," which yield an automatic four runs if the ball reaches the boundary along the ground and six runs if it clears the boundary in the air. Batters may also choose not to run after striking the ball, resulting in a "dot ball" that produces no runs. A batter that runs and ends at the opposite wicket does not face the next ball delivered; instead their partner batter does.

A distinctive institutional feature of T20 cricket is the presence of discrete phases of play with sharply different returns to batting intensity. During the first six overs of an inning, called the "powerplay", only two fielders are permitted outside the 30-yard circle making defense significantly more difficult. For the batting team, this period of play substantially increases the returns to aggressive boundary hitting. After the powerplay, up to five fielders may be positioned outside the circle (fig. 1), reducing the effectiveness of high-intensity shots. Batters enter sequentially, with those higher in the batting order disproportionately exposed to the powerplay. The remaining overs are referred to as the "middle" and "death" overs, during which fielding restrictions remain the same.

3.2 Ramadan and Fasting

Ramadan is a month in the Islamic lunar calendar during which practicing Muslims fast daily from dawn to sunset. Fasting entails abstaining from food, drink, and smoking during daylight hours. Because the Islamic calendar is not synchronized to the Gregorian calendar, Ramadan shifts earlier by approximately 11 days each year, generating quasi-random variation in its overlap with the IPL season across years. In some seasons, Ramadan does not overlap the IPL season at all; in others, it overlaps partially or fully (fig. 2).

Daily fasting has several immediate physiological effects relevant to athletic performance. During the fasting window, individuals may experience dehydration, lower blood glucose,

changes in sleep timing, and reduced caloric intake relative to non-fasting days. These effects can induce fatigue, particularly as the day progresses. IPL matches are typically scheduled in the late afternoon or evening, with many starting before sunset and ending after. For these matches, players who fast may begin the match under fasting conditions and break their fast (iftar) during play or between innings. This timing creates heterogeneous exposure to the physiological shock of fasting both within and across matches.

Ramadan fasting ends with Eid al-Fitr, a holiday marking the conclusion of the fasting month. Eid falls immediately after the last day of Ramadan and fasting ceases thereafter. For IPL seasons in which Ramadan overlaps the tournament, Eid also provides a natural discontinuity after which Muslim players are no longer subject to fasting-related physiological constraints. Not all Muslims fast during Ramadan. However, elite athletes commonly report fasting during competition, and IPL media coverage indicates that many Muslim players continue to fast during the tournament (India Today Web Desk, 2019; Bose, 2021).

3.3 Data

We construct a ball-by-ball panel dataset of Indian Premier League (IPL) matches using JSON data from CricSheet. These data provide detailed information on every ball bowled between 2008 and 2025, including the batter and bowler, runs scored, dismissal type, and match context. As the overlap between IPL season and Ramadan is for a few years we only use 2018-2019 and 2021-2025 for our main results. We merge these data with official IPL match schedules to obtain match dates, venues, and start times. For our DiDs, we only include games that are within the period of Ramadan or the 30 days surrounding it. This restricts each window of observation in each season to at most, a three month period.

Our samples for the difference in RD and DiD estimators are demonstrated in table 1 and table 2 respectively. We note that our estimates are based off of relatively small numbers of Muslim and non-Muslim players although we have a much larger number of at-bats for both groups. At baseline Muslim batters tend to produce fewer runs per ball and are dismissed at higher rates. They tend to also bat further down the order relative to non-Muslim players.

For labour market outcomes, we supplement these data with annual IPL auction records. The IPL operates a distinctive labour market in which teams bid for players each year, subject to rules on player retention and periodic "mega auctions" in which all players re-enter

the bidding pool. These auction data provide information on whether a player was bid on and the final auction price, which we interpret as a proxy for perceived marginal product. We additionally have information on annual player salary for players entering the auction. Other players who are not auctioned negotiate salaries directly with their teams.

3.4 Identifying Muslim Players

Names are associated with group identity across the world and religious practices are an important marker of identity that shape preferences, attitudes and economic outcomes Iyer (2016). Our estimation strategy relies on identifying Muslim and non-Muslim cricket players during and outside of Ramadan. To operationalize this strategy we first designate individuals in our sample of Cricket players as Muslim and non-Muslim. Inferring religion based on individual names is not uncommon in the economic literature (see Sachar et al. (2006), Chaturvedi et al. (2025), Elliott (2009), Oreopoulos (2011) and a substantial number of papers (that usually study electoral behavior and conflict) are dependent on manual classification (Field et al. (2008), Bhalotra et al. (2014), Bhalotra et al. (2009).

We use an algorithm developed by Chaturvedi and Chaturvedi (2024) to identify Muslim and non-Muslim players in IPL cricket records. In contrast to previous work, this categorization of cricket players is based on a character-based machine-learning model that classifies seen and unseen names based on their linguistic origins with high accuracy and spelling variations. In extensions checks we examine whether our results are robust to identifying Muslim players using large language models and by a co-author identifying Muslims directly. These alternate changes in classification do not affect our results.

3.5 Econometric Strategy

Our main identification strategy uses two estimation strategies. First, we exploit the phase transition and rule changes from powerplay to middle overs when comparing Muslim and non-Muslim cricket players during and outside of Ramadan. The forcing variable in our regression discontinuity design is the over index o_b at ball b, with a cutoff at $c = 6$. We define $Post_b = 1\{o_b \geq c\}$. Let $Muslim_i$ denote an indicator that player i is Muslim and $Ramadan_m$ an indicator that match m occurs during Ramadan. Our "treated" group consists of Muslim players during Ramadan or $T_{im} = Muslim_i \times Ramadan_m$. Our pooled difference-in-regression-discontinuities specification is

$$
\begin{aligned}
Y_{imb} = \alpha &+ \theta_0 Post_{imb} + \theta_1(Post_{imb} \times Muslim_i) \\
&+ \theta_2(Post_{imb} \times Ramadan_m) + \theta_3(Post_{imb} \times Muslim_i \times Ramadan_m) \\
&+ \gamma_0 r_{imb} + \gamma_1(r_{imb} \times Post_{imb}) \\
&+ \lambda_{im} + \eta_i + \varepsilon_{imb}. \qquad (27)
\end{aligned}
$$

estimated on a symmetric ± 4 over window around the cutoff. Y_{imb} is runs per ball, a dot-ball indicator, or a boundary indicator. Match-by-inning fixed effects control for pitch conditions and opponent quality. Standard errors are clustered at the match level. The parameter of interest is θ_3, which measures how the post-powerplay discontinuity for Muslims during Ramadan differs from the other three groups. Our identifying assumption is that in absence of Ramadan fasting, the discontinuity between overs 6 and 7 would be the same for Muslim and non-Muslim players which we can observe in our data by examining Muslims outside of Ramadan. We estimate this regression by extending the over bandwidths in robustness checks.

We estimate a complementary difference-in-differences specification using both the onset of Ramadan and the onset of Eid as treatment events. This design compares Muslim and non-Muslim players before, during, and after Ramadan and allows us to test for changes in average productivity over the fasting period. We examine the same primary outcomes, assess whether teams alter batting order or exposure to the powerplay, middle, and death overs during Ramadan. We are also able to triangulate any productivity effects we estimate in the difference in RD to phase in the DiD estimates.

We estimate the following specification:

$$
Y_{ib} = \beta(\text{Muslim}_i \times \text{Ramadan}_t) + \theta \text{Muslim}_i + \lambda \text{Ramadan}_t + \alpha + \delta + \omega + I + \epsilon_i \qquad (28)
$$

where Y_{ib} denotes the outcome for batter i on ball b. The specification absorbs fixed effects for the bowling team (α), venue (δ), time of day (day versus night indicator, ω) and batter (I). As the object of interest is a within-batter reallocation of effort, we include batter fixed effects in the level difference-in-differences, so that the Muslim×Ramadan coefficient is

identified by comparing the same batter fasting versus not, rather than by comparing the selected set of Muslim batters who appear in Ramadan games (disproportionately lower-order players) to those who appear outside it. Standard errors are clustered at the batter level.

As an ancillary test of whether individual-level effort reallocation translates into team-level performance costs, we construct a team-by-match panel and estimate a continuous-treatment DiD specification. The unit of observation is a team-match, yielding two observations per match. We estimate effects on three outcomes: a binary win indicator, run differential, defined as runs scored minus runs conceded, and score percentage, defined as team runs divided by total runs scored in the match. The continuous treatment variable is the number of Muslim players in each team's batting lineup, which we interact with a binary Ramadan indicator equal to one for matches played within the Ramadan window. The estimating equation is:

$$Y_{it} = \alpha + \beta \left(N^{\text{Muslim}} * it \times \text{Ramadan} * t \right) + \gamma N^{\text{Muslim}} * it + \delta, \text{Ramadan} * t + \phi_i + \varepsilon_{it}. \quad (29)$$

where ϕ_i is a team fixed effect and standard errors are clustered at the team level. We additionally estimate the specification separately for day and night matches, since fasting is active only during day games and so we expect stronger effects then.

Small Sample Robustness Checks. Although our estimating samples contain tens of thousands of deliveries, treatment varies across a small number of individuals, and it varies at a level coarser than the matches at which we cluster. Throughout, a batter is a player-franchise unit: a player appearing for more than one franchise is counted once per franchise, consistent with the fixed effects in our specifications. Twenty-two Muslim batters appear in the regression-discontinuity window, of whom fourteen bat during Ramadan and identify the triple-difference coefficient. In the level difference-in-differences, the within-batter interaction is identified by the thirty-four Muslim batters observed both during and outside Ramadan; the death-overs reallocation, which requires within-batter variation in the final phase, rests on the twenty-nine such batters who appear there in both periods. Since treatment is assigned across so few players analytic cluster-robust standard errors can understate the standard error. We therefore do not rely on analytic standard errors alone, and report three additional forms of evidence alongside our primary estimates.

First, we base inference on randomization tests. Holding fixed the number of treated batters, we reassign Muslim status at random across batters, re-estimate the full specifica-

tion, and repeat this 499 times, comparing the observed estimate to the resulting placebo distribution. Muslim batters occupy systematically lower positions in the order and so we implement a stratified version in which placebo assignments match the observed batting-order distribution, so that the placebo players face comparable exposure to each phase of the innings.

Second, we assess whether any single unit drives our results by re-estimating each specification while omitting one batter, and one season at a time. We report the full range of the resulting point estimates rather than only their average, and note explicitly where any omission affects statistical significance. An estimate that changed sign or lost its interpretation under a single omission would indicate that the result reflected the behaviour of one player or one Ramadan-season overlap rather than a systematic response to fasting.

Third, we report standard errors under alternative clustering and resampling schemes. Treatment is assigned at the player level while our discontinuity design compares deliveries within matches so we report analytic standard errors clustered at the match level, at the batter level, and two-way by match and batter, together with pairs-cluster bootstrap standard errors that resample matches and, separately, resample batters. Since the treated batters are far fewer than the matches, batter-level resampling is the most demanding of these procedures, and we adopt it as our conservative benchmark when bounding the aggregate effect.

3.6 Extensions and Robustness Checks

We conduct a series of extensions and robustness checks that test additional implications of our effort reallocation framework and assess alternative behavioral and strategic margins. Together, these analyses examine whether reallocation of effort in response to predictable physiological constraints appears along dimensions other than within-match scoring profiles.

Time of day. The effects of Ramadan should depend on when a match is played and IPL matches occur either in the afternoon or in the evening. In afternoon matches, players have been fasting since sunrise, whereas in evening matches fasting typically ends before play begins. Using the same difference-in-regression-discontinuity design as in our main analysis, we estimate effects separately by time of day. We are able to observe game start times but not times that phases of a game occur. However, consistent with the physiological mechanism, we expect stronger reallocation of effort in matches with start times in the afternoon and

attenuated effects in matches that begin in the evening.

Team strategy and the coin toss. If there are consequential effects of Ramadan fasting, teams might anticipate the performance declines faced by Muslim players and adjust strategy accordingly. In the IPL, the decision to bat first or second is made by the team that wins a pre-match coin toss, providing a source of quasi-random variation in strategic choice. Batting second is generally advantageous in afternoon matches because temperatures fall and fatigue accumulates later in the game. We examine whether teams with more Muslim players are less likely to choose to bat first during Ramadan, particularly in day matches. Restricting the sample to toss winners, we estimate the following linear probability model:

$$\text{BatFirst}_m = \beta(\text{Muslims}_m \times \text{Ramadan}_t) + \theta\text{Muslims}_m + \lambda\text{Ramadan}_t + \gamma + \delta + \epsilon_m \quad (30)$$

where BatFirst_m is an indicator for whether team m chooses to bat first, Muslims_m is the number of Muslim players on the team, and Ramadan_t indicates whether match t occurs during Ramadan. The specification includes year fixed effects (γ_y) and venue fixed effects (δ_v). Standard errors are clustered at the match level. We expect $\beta < 0$ in afternoon matches and no systematic effect in evening matches, when fasting has typically ended and temperatures are lower.

Impact substitutes. The IPL introduced an impact substitute rule beginning with the 2023 season, allowing each team to replace one player mid-match; the substituted player may no longer bat or bowl, while the incoming player assumes a full role. We exploit this rule to test a complementary mechanism: if fasting imposes fatigue costs, team management should respond by substituting Muslim players out of matches at higher rates during Ramadan than outside it. Using match-level impact substitute records for IPL 2023 and 2024, we construct a team-by-match panel restricted to observations in which the impact substitute was deployed. The unit of observation is a team-match, and the primary outcomes are binary indicators for whether a Muslim player was replaced, and whether a Muslim player came in as the substitute.

The baseline specification is a linear probability model regressing the substitution outcome on a Ramadan indicator, with team fixed effects and standard errors clustered at the

match level:

$$Y_{it} = \alpha + \beta, \text{Ramadan} * t + \phi * i + \varepsilon_{it}. \tag{31}$$

where Y_{it} is the Muslim substitution indicator for team i in match t and ϕ_i is a team fixed effect. The sample is restricted to IPL 2023–2024 because the impact substitute rule was not in place in prior seasons.

Labor market outcomes. Finally, we assess whether fasting-related physiological constraints translate into labor market penalties. We analyze player valuations in the IPL auction market and construct a Ramadan Intensity index, $R_t \in [0, 1]$, defined as the fraction of the IPL season overlapping with Ramadan in year t. We estimate the following wage equation:

$$\ln(Price_{it}) = \beta(\text{Muslim}_i \times R_t) + \gamma \text{Muslim}_i + \delta R_t + \mathbf{X}_{it}\Theta + \alpha_i + \lambda_t + \epsilon_{it} \tag{32}$$

where $Price_{it}$ is the final auction price for player i in year t. Player fixed effects (α_i) control for time-invariant ability, and year fixed effects (λ_t) absorb league-wide market conditions, including mega-auction years. We further examine the extensive margin by estimating a linear probability model in which the outcome is an indicator for whether a player is sold at auction.

4 Results

We begin by establishing the effects of fasting on how Muslim players respond to the powerplay boundary hitting advantage in Table 3 using our Dif-in-RD estimator. Non-Muslim batters and Muslim batters outside Ramadan exhibit the conventional drop in productivity at the Powerplay cutoff. This latter observation suggests that our main assumptions in our Dif-in-RD hold. In contrast, Muslim batters during Ramadan do not: the triple-difference estimate indicates an increase of 0.7064 runs per ball at the cutoff ($p = 0.0149$). Converting to runs per over, this corresponds to roughly +4.24 runs per over at the transition. We find similarly signed increases in boundary balls (0.1825), with no detectable effects on dismissals or dot balls.

The anatomy of the effect is informative. Rather than exhibiting a positive discontinuity, Muslim batters during Ramadan simply fail to experience the negative discontinuity that

characterizes their three comparator groups. Figure 3 shows the naive averages of outcomes and demonstrates that non-Muslim batters experience a sharp decline in both scoring and boundaries after over 6, while Ramadan Muslim batters maintain a smooth scoring profile. This pattern suggests strategic smoothing of effort across phases.

To understand the margins of adjustment, we decompose run production in table 4. Nearly the entire DiRD effect is accounted for by boundary hitting, particularly sixes (0.6011 runs per ball). Running-intensive margins (singles, doubles, triples) move in the opposite direction. Together, these results imply that fasting players reduce effort in the powerplay phase to maintain effort in the middle overs.

Our key small-sample robustness checks on these main Dif-in-RD estimates are contained in Table 5. The precision of our estimates are robust to a test of randomization-inference and our headline estimates are larger than all 499 samples of random batsmen. Leave-one-season and leave-one-batter out ranges similarly demonstrate that effects are not driven by outliers.

These Dif-in-RD effects are reinforced by our DiD estimates (Table 6). However where we were unable to make inferences about whether productivity in the death overs with our Dif-in-RD estimates, our DiDs suggest shifts increased runs-per-ball in the middle overs when using the period before the onset of Eid and increased runs-per-ball in the death overs when using the period after Ramadan onset. Consistent with our Dif-in-RDs, our DiDs also implicate boundary hitting as the largest driver of changes in batter productivity. We find little consistent impacts on dismissals or dot balls in our DiD regression results. Overall, we find no difference in player productivity (Panel A) but we do find changes in when productivity occurs (Panel C).

We find similar precision results in our randomization inference and leave-one-out tests. Table 7 shows in declines in the power play in runs per ball and increases in the death overs although these later results are only significant at the 10% level. We find consistent and significant declines in the powerplay boundaries hit and mirror increases in death boundaries.

These changes in when production occurs during a game result in modest changes to how the batting order is organized. We find no change in the number of balls per phase that an individual player is exposed to but we do find mild declines in the batting order of players. Muslim players move up 0.298 places in the batting order as a result of fasting during Ramadan. However, these ordering results become only marginally significant in our

randomization inference tests (Table 7).

We find no evidence that fasting Muslim players have impacts on whether their team wins (Table 8). However, these estimates are not tightly estimated nulls. In Table 9 we show that we can rule out effects on win probability and run differential larger than 27% of a SD.

The physiological mechanism is further supported by heterogeneity across match start times (table 10). Day matches occur while players are fasting; night matches occur after iftar. Although we find effects in both types of game, the DiRD effect is roughly twice as large in day matches (+1.156 vs +0.656 runs per ball) with similar (although insignificant) relative magnitudes for boundaries. This pattern is consistent with fasting-induced fatigue and hydration constraints rather than non-physiological explanations such as direct relgious effects.

Teams also appear to internalize these constraints. Among toss winners, the probability of electing to bat first declines in day matches with higher Muslim representation during Ramadan (table 11), consistent with captains avoiding exposing fasting batters to early-match exertion in heat. No such pattern appears in night matches. This suggests organizational adaptation to physiological shocks.

In later years, fasting also changes whether a Muslim player is substituted out of a game and replaced by an impact substitute. Muslim players are 6.8% more likely to be substituted out of a game during Ramadan as compared to outside of Ramadan. When restricting to teams that have Muslim players, this increases to 10.8%. There are no increases in whether a Muslim player is used as an impact substitute.

Finally, we assess whether the physiological shock carries consequences in the labor market. table 13 shows no evidence that Muslim batters are discounted in years with greater Ramadan overlap; auction prices and sale probabilities remain unchanged once player fixed effects are included. The absence of labor market penalties is consistent with a market that anticipates transitory physiological shocks and views within-match adaptation as sufficient to ensure productivity.

5 Conclusion

This paper studies how high-stakes professionals respond to a predictable physiological constraint. Using ball-by-ball data from the Indian Premier League, we show that Muslim batters during Ramadan do not experience a simple decline in overall productivity. Instead, they appear to reallocate effort within the innings: reducing high-intensity production during the powerplay while preserving or improving performance in later overs.

These findings suggest that physiological shocks need not translate mechanically into lower output when workers can adjust the timing and intensity of effort. In settings where production is divisible and incentives are strong, agents may respond to binding physical constraints by changing how effort is allocated rather than by producing less. More broadly, the results highlight the importance of distinguishing between changes in productive capacity and changes in the allocation of effort when studying health, fatigue, and labor productivity.

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6 Tables and Figures



Figure 1: *Pitch Locations and Fielding Restrictions in IPL T20 Cricket.* The first six overs constitute the powerplay period, during which only two fielders are permitted outside the 30-yard circle. Overs 7–15 are classified as the middle overs, where up to five fielders are allowed outside the circle. The transition from the powerplay to middle overs alters field placement strategies and batting incentives across pitch locations. Note: the positions of fielders are an example for a right hand batsman.

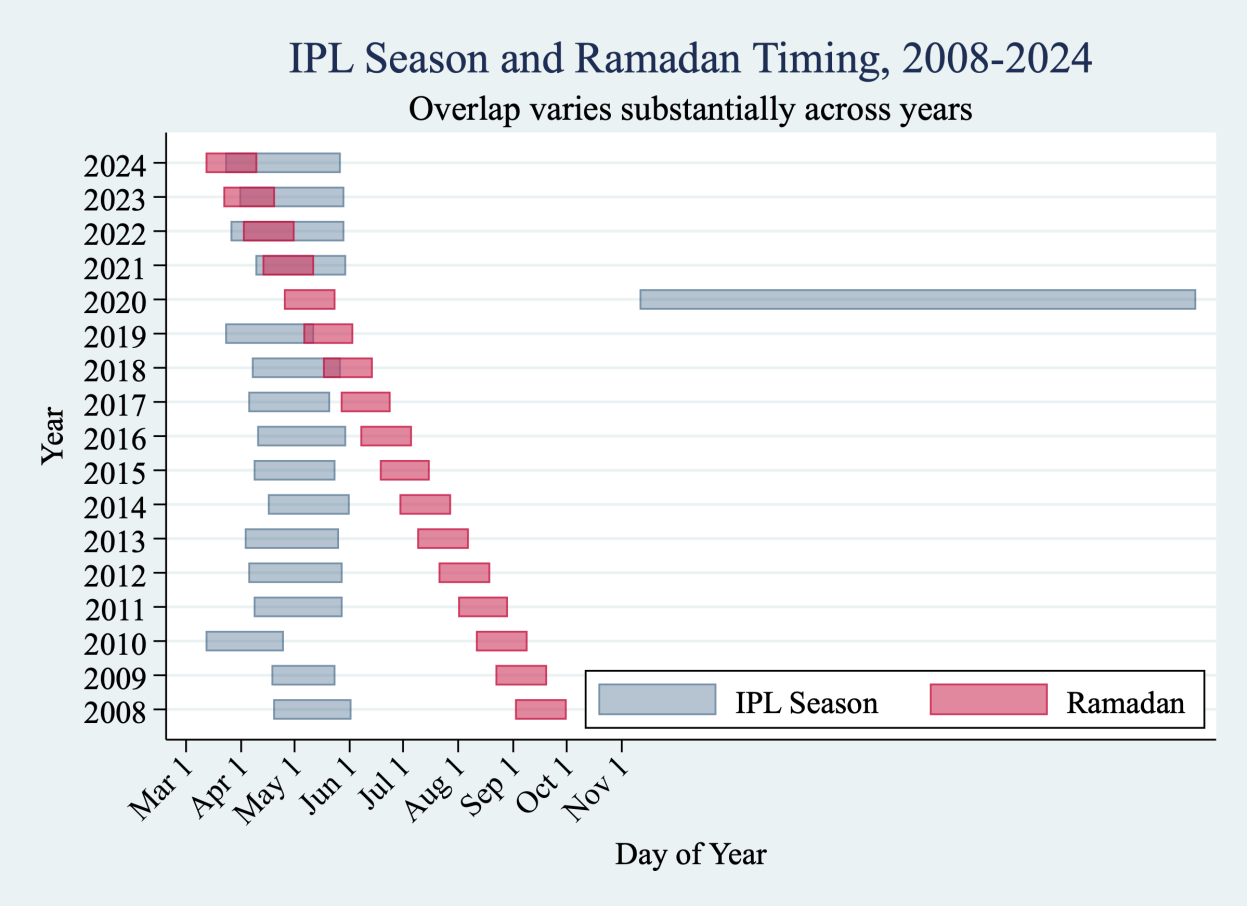


Figure 2: Timing of when Ramadan overlaps with the IPL seasons

Table 1: Descriptive Statistics: Difference-in-Regression-Discontinuities Sample (±24 balls of powerplay cutoff)

	Muslim		Non-Muslim	
	Non-Ramadan	Ramadan	Non-Ramadan	Ramadan
	(1)	(2)	(3)	(4)
Panel A: Sample Composition				
Unique players	17	14	269	209
Unique matches	38	35	209	128
Player-inning obs.	43	39	1567	955
Total balls faced	425	412	19,989	12,120
Panel B: Match Context				
Day games (%)	17.9	14.8	14.6	15.7
Night games (%)	82.1	85.2	85.4	84.3
Powerplay balls (%)	41.2	36.9	51.3	51.6
Middle-over balls (%)	58.8	63.1	48.7	48.4
Death-over balls (%)	0.0	0.0	0.0	0.0
Panel C: Batting Performance (Ball-Level)				
Runs per ball	1.402	1.051	1.342	1.322
	(1.777)	(1.574)	(1.710)	(1.692)
Dot ball (%)	32.2	40.8	31.3	31.2
Boundary (%)	20.7	13.6	18.9	18.3
Four (%)	14.4	9.5	13.3	13.0
Six (%)	6.4	4.1	5.6	5.4
Dismissed (%)	4.00	3.88	3.91	3.94
Odd runs (1 or 3) (%)	36.2	29.4	37.3	37.2
Panel D: Innings-Level Aggregates				
Balls faced per innings	9.88	10.56	12.76	12.69
	(7.47)	(7.72)	(8.25)	(8.26)
Runs per innings	13.86	11.10	17.12	16.77
	(14.25)	(12.94)	(14.60)	(14.33)
Strike rate	125.3	82.3	117.6	113.9
	(62.1)	(52.5)	(64.2)	(61.8)
Batting order	4.51	4.10	3.00	2.94
	(1.61)	(1.68)	(1.58)	(1.51)

Notes: Standard deviations in parentheses. Sample restricted to matches within 30 days of Ramadan/Eid (2018–2025, excluding 2020); day games start before 6pm IST; playoffs excluded. Columns split batters by religion and by whether the match falls in Ramadan. Unique players are counted at the player-franchise level, consistent with the fixed-effects unit used in estimation: a player appearing for more than one franchise is counted once per franchise.

Table 2: Descriptive Statistics: Difference-in-Differences Sample (full innings)

	Muslim		Non-Muslim	
	Non-Ramadan	Ramadan	Non-Ramadan	Ramadan
	(1)	(2)	(3)	(4)
Panel A: Sample Composition				
Unique players	47	47	469	371
Unique matches	114	89	210	128
Player-inning obs.	192	147	2992	1891
Total balls faced	1,952	1,508	48,102	29,678
Panel B: Match Context				
Day games (%)	15.1	16.0	14.6	15.7
Night games (%)	84.9	84.0	85.4	84.3
Powerplay balls (%)	11.2	13.0	32.0	31.6
Middle-over balls (%)	44.9	48.1	46.0	45.4
Death-over balls (%)	44.0	38.9	22.0	23.0
Panel C: Batting Performance (Ball-Level)				
Runs per ball	1.316	1.241	1.370	1.341
	(1.755)	(1.704)	(1.738)	(1.724)
Dot ball (%)	27.9	29.8	28.1	28.7
Boundary (%)	17.2	15.5	18.7	18.0
Four (%)	9.9	8.8	12.2	11.7
Six (%)	7.2	6.6	6.4	6.3
Dismissed (%)	6.97	7.03	4.91	5.04
Odd runs (1 or 3) (%)	35.8	35.1	37.3	37.0
Panel D: Innings-Level Aggregates				
Balls faced per innings	10.17	10.26	16.08	15.69
	(10.37)	(9.75)	(14.14)	(13.78)
Runs per innings	13.38	12.73	22.02	21.05
	(16.43)	(13.69)	(23.18)	(22.11)
Strike rate	111.8	113.3	117.3	114.9
	(78.9)	(87.4)	(69.9)	(71.4)
Batting order	6.67	6.63	4.48	4.60
	(2.22)	(2.47)	(2.61)	(2.66)

Notes: Standard deviations in parentheses. Sample restricted to matches within 30 days of Ramadan/Eid (2018–2025, excluding 2020); day games start before 6pm IST; playoffs excluded. Columns split batters by religion and by whether the match falls in Ramadan. Unique players are counted at the player-franchise level, consistent with the fixed-effects unit used in estimation: a player appearing for more than one franchise is counted once per franchise.

Table 3: Difference-in-Regression-Discontinuities Estimates by Outcome

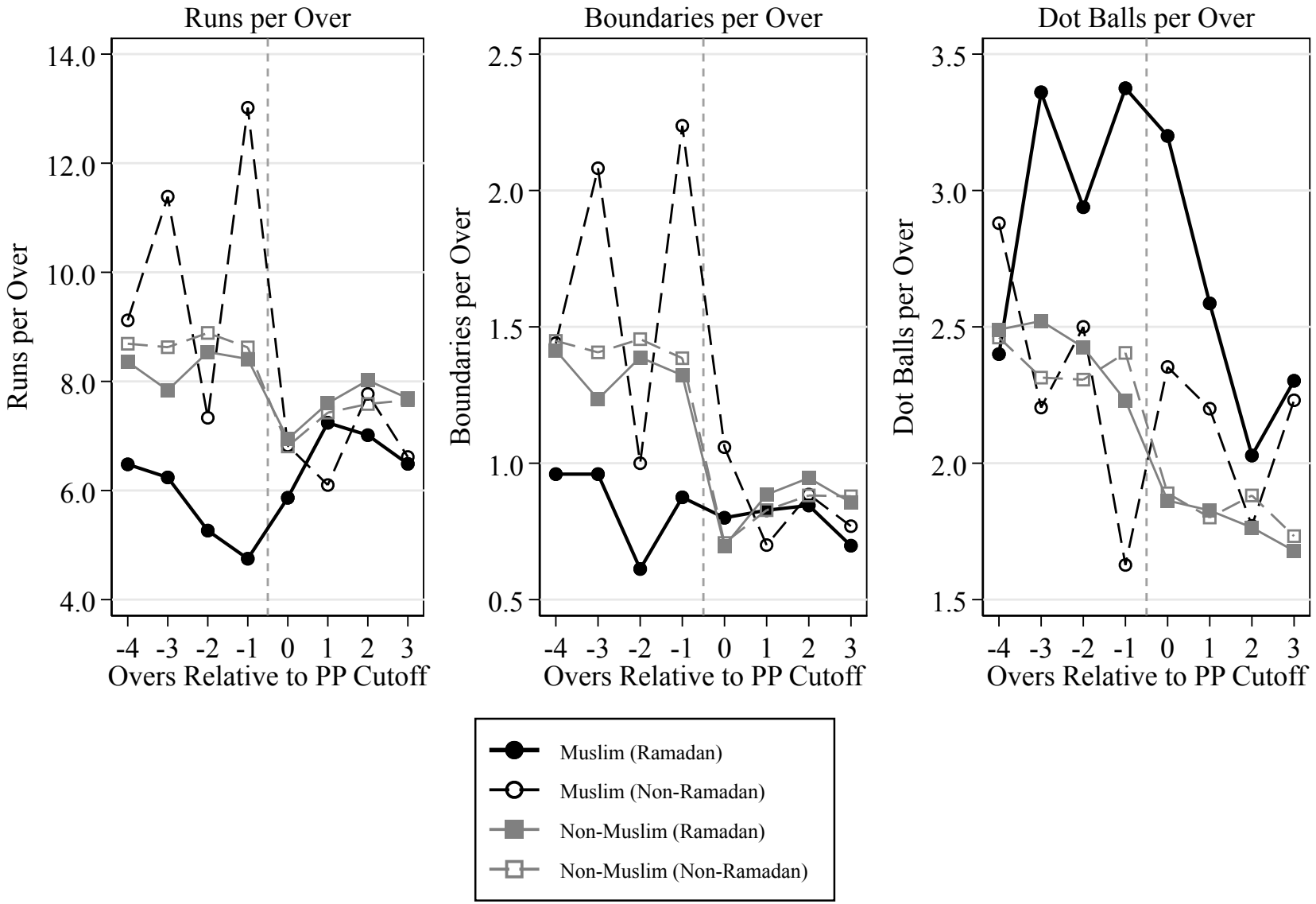
	Runs per ball				Boundary (4+ runs)				Dismissed				Dot ball			
	Coef	SE	p-value	N	Coef	SE	p-value	N	Coef	SE	p-value	N	Coef	SE	p-value	N
Muslim (Ramadan)	0.3039	0.2849	0.2936	412	0.0427	0.0717	0.5557	412	0.0244	0.0546	0.6572	412	-0.0023	0.1010	0.9820	412
Muslim (Other)	-0.8500	0.3835	0.0333	423	-0.1634	0.1003	0.1124	423	-0.0505	0.0365	0.1751	423	0.0666	0.0942	0.4840	423
Non-Muslim (Ramadan)	-0.2120	0.0639	0.0012	12120	-0.0897	0.0149	0.0000	12120	-0.0130	0.0059	0.0285	12120	-0.0652	0.0175	0.0003	12120
Non-Muslim (Other)	-0.3184	0.0493	0.0000	19989	-0.1135	0.0106	0.0000	19989	-0.0003	0.0055	0.9594	19989	-0.0695	0.0158	0.0000	19989
DiRD	0.7064	0.2886	0.0149	32946	0.1245	0.0695	0.0741	32946	-0.0030	0.0308	0.9231	32946	-0.0541	0.0738	0.4639	32946

Notes: Each block of four columns corresponds to a ball-level outcome: runs scored per ball, an indicator for hitting a boundary (4 or more runs), an indicator for dismissal, and an indicator for a dot ball. Entries report regression discontinuity (RD) estimates at the Powerplay cutoff (the last ball of over 5) from local linear models using balls within ±24 deliveries of the cutoff. Rows 1–4 show group-specific RD jumps in the post-Powerplay indicator for Muslim batters during Ramadan, Muslim batters outside Ramadan, non-Muslim batters during Ramadan, and non-Muslim batters outside Ramadan. The DiRD row reports the triple-difference estimate for Post × Muslim × Ramadan from a pooled specification that interacts the RD with Muslim and Ramadan indicators. All models include match-inning as well as linear trends in distance from the cutoff (with interactions by Muslim/Ramadan status in the pooled model). Standard errors are clustered at the match level, and p-values are two-sided. N reports the number of ball-level observations used for each group and outcome. The sample is restricted to matches within ±30 days of Ramadan or Eid and to balls within ±24 of the Powerplay cutoff.

Table 4: Distribution of Run Production Components

	Per Ball	Share of Total (%)	Per Over
Panel A: Running			
Singles (1s)	-.0875309	-12.39048	-.5251852
Twos (2s)	.1068523	15.12554	.6411141
Threes (3s)	-.0116692	-1.65184	-.0700152
Panel B: Boundaries			
Fours (4s)	.0946556	13.39902	.5679336
Sixes (6s)	.6011345	85.09388	3.606807
Sum of Components	.7034423	99.57612	4.220654

Notes: Table 4 decomposes the total DiRD effect on runs per ball into contributions from each run type. Contributions are computed as (DiRD coefficient for run-type indicator) × (run value). "Per Ball" reports the contribution in runs per delivery; "Share of Total" reports the fraction of the total DiRD effect explained by each component; "Per Over" scales the contributions to a six-ball over. Singles, twos, and threes reflect running-based production, while fours and sixes reflect boundary-hitting. Negative shares for running components indicate compensatory behaviour. The sum row verifies that individual contributions sum to the total DiRD effect. All components are estimated using the RDD specification at the Powerplay cutoff, with match-inning and batter fixed effects and linear trends in distance from the cutoff. Standard errors clustered at the match level. Sample restricted to ±24 balls around the cutoff.



Note: Sample restricted to ±24 balls around cutoff. Each panel shows average outcome per over.

Figure 3: Difference in RD effects when transitioning to the Middle Overs

Table 5: Inference and Stability of the Powerplay Discontinuity

DiRD estimate (SE) [analytic p]	0.706 (0.289) [0.015]
Randomization-inference p (placebos $\geq$ observed / repetitions)	0.002 (0/499)
Leave-one-season-out range	[0.531, 0.827]
Leave-one-batter-out range	[0.501, 0.852]

The powerplay difference-in-discontinuities is estimated with match×inning fixed effects. Randomization inference reassigns Muslim status across the 14 Muslim batters in the window (499 repetitions). Leave-one-out specifications drop each season or each Muslim batter and re-estimate the model. Standard errors are clustered at the match level.

Table 6: Difference-in-Differences: Aggregate, Exposure, and Intensity Margins

	Before Eid		After Ramadan		Pooled	
Outcome	ATT	SE	ATT	SE	ATT	SE
Panel A: Aggregate outcomes (all phases)						
Runs per ball	0.255**	(0.119)	-0.082	(0.093)	0.007	(0.081)
Dot ball	-0.071*	(0.039)	0.016	(0.029)	-0.005	(0.026)
Four (4)	0.027	(0.028)	-0.012	(0.018)	-0.003	(0.016)
Six (6)	0.013	(0.024)	-0.004	(0.012)	0.002	(0.011)
Dismissed	0.003	(0.023)	0.007	(0.012)	0.006	(0.011)
Panel B: Exposure and role						
Powerplay share	-0.023	(0.051)	0.056	(0.048)	0.039	(0.041)
Middle share	0.018	(0.108)	-0.003	(0.059)	-0.004	(0.054)
Death share	0.006	(0.076)	-0.053	(0.046)	-0.035	(0.041)
Batting order	0.104	(0.161)	-0.387**	(0.181)	-0.298**	(0.148)
Panel C: Intensity within phase						
Runs/ball: Powerplay	-0.373**	(0.180)	-0.543*	(0.309)	-0.507**	(0.236)
Runs/ball: Middle	0.402**	(0.182)	0.016	(0.123)	0.128	(0.110)
Runs/ball: Death	0.321	(0.301)	0.267*	(0.146)	0.277**	(0.138)
Boundary: Powerplay	-0.037	(0.045)	-0.131*	(0.078)	-0.113*	(0.061)
Boundary: Middle	0.055	(0.047)	-0.006	(0.029)	0.010	(0.026)
Boundary: Death	0.100*	(0.054)	0.064**	(0.032)	0.074**	(0.029)
Dismissed: Powerplay	0.025	(0.054)	-0.060	(0.037)	-0.044	(0.033)
Dismissed: Middle	-0.060*	(0.035)	0.012	(0.021)	-0.007	(0.019)
Dismissed: Death	0.023	(0.055)	0.005	(0.032)	0.010	(0.028)
Dot ball: Powerplay	0.007	(0.097)	0.049	(0.078)	0.026	(0.066)
Dot ball: Middle	-0.083	(0.067)	-0.036	(0.036)	-0.045	(0.036)
Dot ball: Death	0.002	(0.072)	0.002	(0.048)	0.005	(0.042)

Notes: Each cell is the Muslim × Ramadan interaction (ATT) from a separate regression with match-inning and batter fixed effects; SEs clustered at match level. *Panel A* pools all phases. *Panel B* reports exposure and role: the share of a batter's balls in each phase (the three shares partition to one, so their coefficients sum to zero) and batting order (estimated at the batter-innings level, since it is constant within an innings). *Panel C* estimates ball-level outcomes — runs per ball, boundary, dismissal, and dot-ball rates — within each phase subsample; the runs-per-ball rows carry the reallocation (a powerplay loss re-appearing as a death-over gain that ball-weights to the Panel A runs null). Because Ramadan lasts ∼30 days, splitting Panel C by Before/After halves each cell, so the *Pooled* column is the powered estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Inference and Stability of the Reallocation: Powerplay Decline and Death Gain

Margin	Estimate (SE)	Analytic p	RI p	Leave-one-season	Leave-one-batter
Panel A: Aggregate (all balls)					
Runs per ball	0.007 (0.081)	0.931	0.920	[-0.032, 0.108]	[-0.018, 0.055]
Boundary rate	-0.001 (0.017)	0.936	0.914	[-0.015, 0.019]	[-0.007, 0.008]
Panel B: Exposure					
Powerplay ball-share	0.039 (0.041)	0.342	0.310	[0.024, 0.051]	[-0.001, 0.062]
Batting order	-0.298 (0.148)	0.045	0.092	[-0.352, -0.228]	[-0.357, -0.200]
Panel C: Intensity by phase (runs per ball)					
Runs/ball: powerplay	-0.507 (0.236)	0.032	0.012	[-0.858, -0.258]	[-0.703, -0.370]
Runs/ball: death	0.277 (0.138)	0.046	0.054	[0.206, 0.345]	[0.210, 0.378]
Panel D: Boundary reallocation (by phase)					
Boundary: powerplay	-0.113 (0.061)	0.066	0.014	[-0.209, -0.039]	[-0.148, -0.083]
Boundary: death	0.074 (0.029)	0.011	0.012	[0.058, 0.097]	[0.063, 0.092]
Panel E: Powerplay detail					
Dot ball: powerplay	0.026 (0.066)	0.693	0.572	[-0.010, 0.092]	[-0.027, 0.061]

Each row is the Muslim×Ramadan interaction from a level difference-in-differences with match×inning *and batter* fixed effects (within-batter), standard errors clustered at the match level. Aggregate and exposure rows use all deliveries; intensity and boundary rows condition on the stated phase; batting order is estimated at the batter-innings level. Analytic p is the cluster-robust two-sided value. Randomization-inference p reassigns Muslim status across the batters in each sample (499 reps, uniform). Leave-one-out columns report the range of the point estimate dropping each season, and each Muslim batter observed in both Ramadan and non-Ramadan games (6 such batters in the powerplay, 29 in the death overs).

Table 8: Team Productivity During Ramadan

	Pooled			By time of day	
	Win	Run diff.	Score %	Win (day)	Win (night)
N Muslim players	-0.135***	-5.292**	-0.009**	-0.152	-0.139***
	(0.037)	(1.776)	(0.003)	(0.136)	(0.033)
Ramadan	0.022	1.682	0.002	0.039	0.013
	(0.062)	(2.644)	(0.004)	(0.155)	(0.071)
Ramadan × N Muslims	-0.048	-2.507	-0.003	-0.082	-0.042
	(0.050)	(2.597)	(0.004)	(0.164)	(0.051)
Observations	670	670	670	98	572
Dep. var. mean (non-Ramadan)	0.500	0.000	0.500	0.500	0.500

Notes: Unit is team × match (two observations per match), with team fixed effects and standard errors clustered at the team level. Sample: IPL 2018–2025 (excluding 2020), within 30 days of Ramadan/Eid; day matches start before 6pm IST (fasting active); abandoned and tied matches excluded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Team-Level Effects: Point Estimates and Equivalence Bounds

Outcome	ATT	SE	Equiv. bound	MDE (80%)	% of SD
Win probability	-0.0482	(0.0501)	±0.1376	0.1529	27.5%
Run differential	-2.5074	(2.5971)	±7.1362	7.9249	27.1%
Score share	-0.0033	(0.0038)	±0.0101	0.0117	25.0%

The sample is a team-match panel. Treatment is the number of Muslim players interacted with Ramadan. All specifications include team fixed effects, with standard errors clustered at the team level (13 clusters). ATT is the per-Muslim-player Ramadan effect. The equivalence bound is the smallest symmetric effect rejected using a one-sided 5% test. MDE is the minimum effect detectable with 80% power at the 5% significance level.

Table 10: Mechanism Test: Day vs Night Games

	Runs per Ball		Boundary (4+ runs)	
	(1) Day Games	(2) Night Games	(3) Day Games	(4) Night Games
DiRD: Post × Muslim × Ramadan	1.156*	0.656**	0.177	0.131
	(0.639)	(0.331)	(0.134)	(0.081)
Observations	4,949	27,997	4,949	27,997
Adjusted R^2	0.009	0.018	0.020	0.026

Notes: Day games start before 6pm (fasting). Night games start after 6pm (fed). DiRD is the triple-difference estimator at the powerplay cutoff. Match-inning and batter FE included. SEs clustered at match level. * p<0.10, ** p<0.05, *** p<0.01.

Table 11: Strategic Toss Decisions: Do Captains Protect Muslim Players?

	Chose to Bat First (=1)	
	(1) Day Games	(2) Night Games
# Muslims × Ramadan	-0.325*	0.014
	(0.183)	(0.082)
# Muslim Players	0.107	-0.061
	(0.175)	(0.046)
Ramadan Period	0.201	-0.112
	(0.282)	(0.104)
Observations	30	151
Adjusted R^2	-0.133	0.131

Notes: This table tests whether team captains strategically avoid batting first when they have more Muslim players during Ramadan. The dependent variable equals 1 if the team that won the toss chose to bat first (inning 1), and 0 if they chose to field first (inning 2). The sample is restricted to teams that won the toss. The key coefficient is '# Muslims × Ramadan': a negative coefficient in day games indicates captains are less likely to bat first (exposing players to fasting in the heat) when they have more Muslim players during Ramadan. All models include year and venue fixed effects. Standard errors clustered at match level. Sample: ±30 days around Ramadan/Eid. * p<0.10, ** p<0.05, *** p<0.01.

Table 12: Impact-substitute usage during Ramadan

	Muslim OUT	Muslim IN	Muslim OUT†
Ramadan	0.068**	0.016	0.102**
	(0.030)	(0.029)	(0.048)
Observations	292	292	117

Notes: Outcome = 1 if the impact substitute involves a Muslim player (columns 1 and 3: the player replaced; column 2: the player brought on). Unit of observation: team × match, IPL 2023–2024 (the Impact-Player rule exists only from 2023, and both seasons overlap Ramadan), restricted to matches in which the substitution was used. All specifications include team fixed effects; standard errors clustered at the match level. †Column 3 restricts to team-matches with ≥ 1 Muslim player. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Labor Market Outcomes: Auction Prices and Sale Probability

	Log Auction Price		Sold at Auction	
	(1) Pooled OLS	(2) Player FE	(3) Pooled OLS	(4) Player FE
Muslim × Ramadan Share	-0.807	0.853	0.167	0.072
	(0.899)	(0.680)	(0.116)	(0.148)
Mega Auction Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Player FE	No	Yes	No	Yes
Observations	1,213	1,105	1,376	1,245
Adjusted R^2	0.031	0.599	0.229	0.312

Notes: This table examines whether Muslim players face labor market penalties in years with greater Ramadan overlap. Columns (1)-(2) use log auction price as the dependent variable; columns (3)-(4) use an indicator for being sold at auction. The key coefficient is Muslim × Ramadan Share, where Ramadan Share is the fraction of the IPL season overlapping with Ramadan (0–1). Negative coefficients would indicate Muslim players are discounted or less likely to be sold in high-Ramadan years. Columns (1) and (3) include year fixed effects; columns (2) and (4) include both year and player fixed effects. Mega Auction is an indicator for years 2018 and 2022. Standard errors clustered at the player level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Difference-in-Regression-Discontinuities Estimates by Outcome, across name classifiers

	Runs per ball			Boundary (4+)			Dismissed			Dot ball			
	Coef	SE	p	Coef	SE	p	Coef	SE	p	Coef	SE	p	N
Panel A: gpt													
Muslim (Ramadan)	0.304	(0.290)	0.301	0.043	(0.073)	0.562	0.024	(0.055)	0.662	-0.002	(0.103)	0.982	412
Muslim (Other)	-0.760	(0.385)	0.056	-0.150	(0.099)	0.139	-0.057	(0.033)	0.096	0.034	(0.098)	0.729	452
Non-Muslim (Ramadan)	-0.217	(0.065)	0.001	-0.092	(0.015)	0.000	-0.014	(0.006)	0.024	-0.064	(0.018)	0.000	12,115
Non-Muslim (Other)	-0.335	(0.049)	0.000	-0.116	(0.010)	0.000	-0.002	(0.006)	0.737	-0.066	(0.016)	0.000	19,956
DiRD	0.972	(0.297)	0.001	0.181	(0.069)	0.010	-0.012	(0.033)	0.719	-0.094	(0.078)	0.226	32,941
Panel B: gemini													
Muslim (Ramadan)	0.304	(0.290)	0.301	0.043	(0.073)	0.562	0.024	(0.055)	0.662	-0.002	(0.103)	0.982	412
Muslim (Other)	-0.760	(0.385)	0.056	-0.150	(0.099)	0.139	-0.057	(0.033)	0.096	0.034	(0.098)	0.729	452
Non-Muslim (Ramadan)	-0.217	(0.065)	0.001	-0.092	(0.015)	0.000	-0.014	(0.006)	0.024	-0.064	(0.018)	0.000	12,115
Non-Muslim (Other)	-0.335	(0.049)	0.000	-0.116	(0.010)	0.000	-0.002	(0.006)	0.737	-0.066	(0.016)	0.000	19,956
DiRD	0.972	(0.297)	0.001	0.181	(0.069)	0.010	-0.012	(0.033)	0.719	-0.094	(0.078)	0.226	32,941
Panel C: claude													
Muslim (Ramadan)	0.042	(0.315)	0.895	-0.016	(0.080)	0.845	0.057	(0.068)	0.407	0.112	(0.113)	0.330	296
Muslim (Other)	-0.939	(0.414)	0.031	-0.231	(0.108)	0.041	-0.045	(0.036)	0.225	0.003	(0.081)	0.971	352
Non-Muslim (Ramadan)	-0.210	(0.065)	0.002	-0.090	(0.015)	0.000	-0.014	(0.006)	0.023	-0.065	(0.018)	0.000	12,231
Non-Muslim (Other)	-0.333	(0.048)	0.000	-0.115	(0.010)	0.000	-0.002	(0.006)	0.694	-0.065	(0.015)	0.000	20,056
DiRD	0.814	(0.361)	0.025	0.172	(0.084)	0.040	-0.008	(0.037)	0.834	-0.074	(0.089)	0.405	32,941
Panel D: claude2													
Muslim (Ramadan)	0.304	(0.290)	0.301	0.043	(0.073)	0.562	0.024	(0.055)	0.662	-0.002	(0.103)	0.982	412
Muslim (Other)	-0.760	(0.385)	0.056	-0.150	(0.099)	0.139	-0.057	(0.033)	0.096	0.034	(0.098)	0.729	452
Non-Muslim (Ramadan)	-0.217	(0.065)	0.001	-0.092	(0.015)	0.000	-0.014	(0.006)	0.024	-0.064	(0.018)	0.000	12,115
Non-Muslim (Other)	-0.335	(0.049)	0.000	-0.116	(0.010)	0.000	-0.002	(0.006)	0.737	-0.066	(0.016)	0.000	19,956
DiRD	0.972	(0.297)	0.001	0.181	(0.069)	0.010	-0.012	(0.033)	0.719	-0.094	(0.078)	0.226	32,941
Panel E: manual													
Muslim (Ramadan)	0.304	(0.290)	0.301	0.043	(0.073)	0.562	0.024	(0.055)	0.662	-0.002	(0.103)	0.982	412
Muslim (Other)	-0.902	(0.393)	0.028	-0.181	(0.103)	0.088	-0.059	(0.034)	0.092	0.072	(0.098)	0.471	423
Non-Muslim (Ramadan)	-0.217	(0.065)	0.001	-0.092	(0.015)	0.000	-0.014	(0.006)	0.024	-0.064	(0.018)	0.000	12,115
Non-Muslim (Other)	-0.333	(0.049)	0.000	-0.116	(0.010)	0.000	-0.002	(0.006)	0.726	-0.067	(0.016)	0.000	19,985
DiRD	0.983	(0.303)	0.001	0.182	(0.071)	0.011	-0.003	(0.032)	0.924	-0.097	(0.079)	0.222	32,941

Notes: Each panel reports estimates in Table 3 for different name classifiers for robustness (gpt, gemini, claude, claude2, manual). Rows 1–4 are group-specific RD jumps in the post-powerplay indicator (last ball of over 5) from local linear models within ±24 deliveries; the DiRD row is the pooled Post × Muslim × Ramadan triple difference. Match-inning and batter fixed effects; linear trend in distance from cutoff interacted with Muslim/Ramadan; sample within ±30 days of Ramadan/Eid; standard errors clustered at the match level; p is the two-sided value. The N column reports ball-level observations (constant across the four outcomes within a row).

Table 15: Difference-in-Differences: Aggregate, Exposure, and Intensity Margins across name classifiers — **Before Eid**

	gpt	gemini	claude	claude2	manual
Panel A: Aggregate outcomes (all phases)					
Runs per ball	0.255** (0.119)	0.255** (0.119)	0.255** (0.120)	0.255** (0.119)	0.255** (0.119)
Dot ball	-0.071* (0.039)	-0.071* (0.039)	-0.069* (0.040)	-0.071* (0.039)	-0.071* (0.039)
Four (4)	0.027 (0.028)	0.027 (0.028)	0.027 (0.028)	0.027 (0.028)	0.027 (0.028)
Six (6)	0.013 (0.024)	0.013 (0.024)	0.014 (0.024)	0.013 (0.024)	0.013 (0.024)
Dismissed	0.003 (0.023)	0.003 (0.023)	0.005 (0.023)	0.003 (0.023)	0.003 (0.023)
Panel B: Exposure and role					
Powerplay share	0.032 (0.042)	0.032 (0.042)	0.036 (0.043)	0.032 (0.042)	0.032 (0.042)
Middle share	-0.127 (0.099)	-0.127 (0.099)	-0.126 (0.101)	-0.127 (0.099)	-0.127 (0.099)
Death share	0.094 (0.070)	0.094 (0.070)	0.090 (0.071)	0.094 (0.070)	0.094 (0.070)
Batting order	0.104 (0.161)	0.104 (0.161)	0.127 (0.164)	0.104 (0.161)	0.104 (0.161)
Panel C: Intensity within phase					
Runs/ball: Powerplay	-0.373** (0.180)	-0.373** (0.180)	-0.373** (0.180)	-0.373** (0.180)	-0.373** (0.180)
Runs/ball: Middle	0.402** (0.182)	0.402** (0.182)	0.402** (0.182)	0.402** (0.182)	0.402** (0.182)
Runs/ball: Death	0.321 (0.301)	0.321 (0.301)	0.322 (0.306)	0.321 (0.301)	0.321 (0.301)
Boundary: Powerplay	-0.037 (0.045)	-0.037 (0.045)	-0.037 (0.045)	-0.037 (0.045)	-0.037 (0.045)
Boundary: Middle	0.055 (0.047)	0.055 (0.047)	0.055 (0.047)	0.055 (0.047)	0.055 (0.047)
Boundary: Death	0.100* (0.054)	0.100* (0.054)	0.102* (0.054)	0.100* (0.054)	0.100* (0.054)
Dismissed: Powerplay	0.025 (0.054)	0.025 (0.054)	0.025 (0.054)	0.025 (0.054)	0.025 (0.054)
Dismissed: Middle	-0.060* (0.035)	-0.060* (0.035)	-0.060* (0.035)	-0.060* (0.035)	-0.060* (0.035)
Dismissed: Death	0.023 (0.055)	0.023 (0.055)	0.030 (0.055)	0.023 (0.055)	0.023 (0.055)
Dot ball: Powerplay	0.007 (0.097)	0.007 (0.097)	0.007 (0.097)	0.007 (0.097)	0.007 (0.097)
Dot ball: Middle	-0.083 (0.067)	-0.083 (0.067)	-0.083 (0.067)	-0.083 (0.067)	-0.083 (0.067)
Dot ball: Death	0.002 (0.072)	0.002 (0.072)	0.010 (0.073)	0.002 (0.072)	0.002 (0.072)

Notes: Sample window: Before Eid. Columns are name classifiers. Each cell is the Muslim × Ramadan ATT with the standard error in parentheses, from a separate regression with match-inning and batter fixed effects; SEs clustered at the match level. Panel A pools all phases; Panel B is at the batter-innings level; Panel C conditions on the stated phase. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Difference-in-Differences: Aggregate, Exposure, and Intensity Margins across name classifiers — **After Ramadan**

	gpt	gemini	claude	claude2	manual
Panel A: Aggregate outcomes (all phases)					
Runs per ball	-0.082 (0.093)	-0.082 (0.093)	-0.002 (0.105)	-0.082 (0.093)	-0.082 (0.093)
Dot ball	0.016 (0.029)	0.016 (0.029)	-0.010 (0.033)	0.016 (0.029)	0.016 (0.029)
Four (4)	-0.012 (0.018)	-0.012 (0.018)	-0.005 (0.021)	-0.012 (0.018)	-0.012 (0.018)
Six (6)	-0.004 (0.012)	-0.004 (0.012)	0.003 (0.013)	-0.004 (0.012)	-0.004 (0.012)
Dismissed	0.007 (0.012)	0.007 (0.012)	0.003 (0.014)	0.007 (0.012)	0.007 (0.012)
Panel B: Exposure and role					
Powerplay share	0.054 (0.038)	0.054 (0.038)	0.050 (0.041)	0.054 (0.038)	0.054 (0.038)
Middle share	-0.030 (0.052)	-0.030 (0.052)	-0.021 (0.053)	-0.030 (0.052)	-0.030 (0.052)
Death share	-0.024 (0.045)	-0.024 (0.045)	-0.029 (0.049)	-0.024 (0.045)	-0.024 (0.045)
Batting order	-0.387** (0.181)	-0.387** (0.181)	-0.286 (0.198)	-0.387** (0.181)	-0.387** (0.181)
Panel C: Intensity within phase					
Runs/ball: Powerplay	-0.543* (0.309)	-0.543* (0.309)	-0.612 (0.465)	-0.543* (0.309)	-0.543* (0.309)
Runs/ball: Middle	0.016 (0.123)	0.016 (0.123)	0.079 (0.141)	0.016 (0.123)	0.016 (0.123)
Runs/ball: Death	0.267* (0.146)	0.267* (0.146)	0.319** (0.153)	0.267* (0.146)	0.267* (0.146)
Boundary: Powerplay	-0.131* (0.078)	-0.131* (0.078)	-0.186 (0.114)	-0.131* (0.078)	-0.131* (0.078)
Boundary: Middle	-0.006 (0.029)	-0.006 (0.029)	0.011 (0.031)	-0.006 (0.029)	-0.006 (0.029)
Boundary: Death	0.064** (0.032)	0.064** (0.032)	0.069** (0.033)	0.064** (0.032)	0.064** (0.032)
Dismissed: Powerplay	-0.060 (0.037)	-0.060 (0.037)	-0.052 (0.052)	-0.060 (0.037)	-0.060 (0.037)
Dismissed: Middle	0.012 (0.021)	0.012 (0.021)	0.013 (0.022)	0.012 (0.021)	0.012 (0.021)
Dismissed: Death	0.005 (0.032)	0.005 (0.032)	-0.003 (0.033)	0.005 (0.032)	0.005 (0.032)
Dot ball: Powerplay	0.049 (0.078)	0.049 (0.078)	0.060 (0.107)	0.049 (0.078)	0.049 (0.078)
Dot ball: Middle	-0.036 (0.036)	-0.036 (0.036)	-0.070* (0.041)	-0.036 (0.036)	-0.036 (0.036)
Dot ball: Death	0.002 (0.048)	0.002 (0.048)	-0.017 (0.049)	0.002 (0.048)	0.002 (0.048)

Notes: Sample window: After Ramadan. Columns are name classifiers. Each cell is the Muslim × Ramadan ATT with the standard error in parentheses, from a separate regression with match-inning and batter fixed effects; SEs clustered at the match level. Panel A pools all phases; Panel B is at the batter-innings level; Panel C conditions on the stated phase. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Difference-in-Differences: Aggregate, Exposure, and Intensity Margins across name classifiers — **Pooled**

	gpt	gemini	claude	claude2	manual
Panel A: Aggregate outcomes (all phases)					
Runs per ball	0.007 (0.081)	0.007 (0.081)	0.083 (0.088)	0.007 (0.081)	0.007 (0.081)
Dot ball	-0.005 (0.026)	-0.005 (0.026)	-0.027 (0.029)	-0.005 (0.026)	-0.005 (0.026)
Four (4)	-0.003 (0.016)	-0.003 (0.016)	0.003 (0.018)	-0.003 (0.016)	-0.003 (0.016)
Six (6)	0.002 (0.011)	0.002 (0.011)	0.008 (0.012)	0.002 (0.011)	0.002 (0.011)
Dismissed	0.006 (0.011)	0.006 (0.011)	0.004 (0.013)	0.006 (0.011)	0.006 (0.011)
Panel B: Exposure and role					
Powerplay share	0.052* (0.031)	0.052* (0.031)	0.050 (0.033)	0.052* (0.031)	0.052* (0.031)
Middle share	-0.052 (0.047)	-0.052 (0.047)	-0.050 (0.048)	-0.052 (0.047)	-0.052 (0.047)
Death share	-0.001 (0.039)	-0.001 (0.039)	-0.001 (0.042)	-0.001 (0.039)	-0.001 (0.039)
Batting order	-0.298** (0.148)	-0.298** (0.148)	-0.194 (0.158)	-0.298** (0.148)	-0.298** (0.148)
Panel C: Intensity within phase					
Runs/ball: Powerplay	-0.507** (0.236)	-0.507** (0.236)	-0.485 (0.297)	-0.507** (0.236)	-0.507** (0.236)
Runs/ball: Middle	0.128 (0.110)	0.128 (0.110)	0.193 (0.120)	0.128 (0.110)	0.128 (0.110)
Runs/ball: Death	0.277** (0.138)	0.277** (0.138)	0.321** (0.145)	0.277** (0.138)	0.277** (0.138)
Boundary: Powerplay	-0.113* (0.061)	-0.113* (0.061)	-0.124 (0.080)	-0.113* (0.061)	-0.113* (0.061)
Boundary: Middle	0.010 (0.026)	0.010 (0.026)	0.026 (0.027)	0.010 (0.026)	0.010 (0.026)
Boundary: Death	0.074** (0.029)	0.074** (0.029)	0.079*** (0.030)	0.074** (0.029)	0.074** (0.029)
Dismissed: Powerplay	-0.044 (0.033)	-0.044 (0.033)	-0.030 (0.043)	-0.044 (0.033)	-0.044 (0.033)
Dismissed: Middle	-0.007 (0.019)	-0.007 (0.019)	-0.009 (0.020)	-0.007 (0.019)	-0.007 (0.019)
Dismissed: Death	0.010 (0.028)	0.010 (0.028)	0.005 (0.029)	0.010 (0.028)	0.010 (0.028)
Dot ball: Powerplay	0.026 (0.066)	0.026 (0.066)	0.025 (0.079)	0.026 (0.066)	0.026 (0.066)
Dot ball: Middle	-0.045 (0.036)	-0.045 (0.036)	-0.072* (0.038)	-0.045 (0.036)	-0.045 (0.036)
Dot ball: Death	0.005 (0.042)	0.005 (0.042)	-0.009 (0.043)	0.005 (0.042)	0.005 (0.042)

Notes: Sample window: Pooled. Columns are name classifiers. Each cell is the Muslim × Ramadan ATT with the standard error in parentheses, from a separate regression with match-inning and batter fixed effects; SEs clustered at the match level. Panel A pools all phases; Panel B is at the batter-innings level; Panel C conditions on the stated phase. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.